

# SENTINELWIRE™ v.1.1

## A LEGAL AND EVIDENTIARY ASSESSMENT OF **SENTINELWIRE™ v.1.1**

A Wireshark-Driven Network Investigation Framework



(Academic & Regulatory Evaluation)

**A Legal and Evidentiary Assessment  
of  
SentinelWire™ v1.1**

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**December 2025**

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## Executive Summary

### Legal and Evidentiary Assessment of SentinelWire™ v1.1

#### A Wireshark-Driven Network Investigation Framework

##### 1. Purpose and Context

This executive summary provides a high-level legal, regulatory, and evidentiary assessment of SentinelWire™ v1.1, a structured, analyst-driven framework for network investigation and threat detection. Designed for Security Operations Centers (SOCs), incident responders, and forensic teams, SentinelWire™ addresses a critical gap in modern cybersecurity operations: the need for legally defensible, transparent, and compliance-aware methodologies in network investigations.

The analysis evaluates SentinelWire™ not merely as a technical tool, but as a structured investigative methodology with significant implications for data protection, regulatory compliance, digital forensics, and evidentiary reliability.

##### 2. What SentinelWire™ Is (and Is Not)

| What It Is                                                        | What It Is Not                                     |
|-------------------------------------------------------------------|----------------------------------------------------|
| A human-in-the-loop, explainable investigation framework          | An automated detection or decision-making system   |
| Built on Wireshark, a court-recognized and industry-standard tool | A SIEM, IDS/IPS, or endpoint detection platform    |
| Focused on behavioral analysis of network metadata                | A black-box analytics or AI-driven verdict engine  |
| Designed to produce auditable, traceable, and documented outputs  | A system that modifies or alters original evidence |

This distinction is legally significant. Courts and regulators assess the methodology and process behind investigative conclusions, not just the software itself.

##### 3. Key Legal Findings (High-Level)

###### 3.1 Legal Characterization

SentinelWire™ is best classified as an investigative methodology rather than an autonomous monitoring tool. This positions it favorably under legal standards that prioritize transparency, reproducibility, and human oversight.

### **3.2 Explainability and Accountability**

All analytical outputs are derived from explicit, inspectable rules applied to packet-level data. This aligns with modern legal principles of algorithmic accountability and the “right to an explanation”, particularly under European regulatory frameworks such as the proposed AI Act.

## **4. Data Protection and Privacy Assessment**

### **4.1 Nature of the Data**

SentinelWire™ processes network metadata only including IP addresses, DNS queries, TLS SNI, and timing data. It does not decrypt content or inspect application payloads, significantly limiting privacy impact.

### **4.2 GDPR Compliance Position**

The framework supports lawful deployment under:

- **Article 6(1)(f) GDPR** – Legitimate interests in network security
- **Article 6(1)(c) GDPR** – Compliance with legal security obligations (e.g., NIS2)
- **Article 32 GDPR** – Security of processing

Its design embodies data minimization, purpose limitation, and proportionality, reinforced by short, analyst-controlled capture windows (30–60 seconds).

## **5. Cybersecurity Regulation & Standards Alignment**

### **5.1 NIS2 Directive**

SentinelWire™ directly supports incident detection, analysis, and reporting obligations under the NIS2 Directive, enabling early identification of security events and structured incident handling.

### **5.2 ISO/IEC 27001 & 27002**

The framework aligns with key controls including:

- **A.8.15 Logging and A.8.16 Monitoring**
- **A.5.24 Incident management**
- **A.8.8 Management of technical vulnerabilities**

Its documentation-centric approach provides audit-ready outputs and demonstrates a systematic security posture.

## **6. Evidentiary Value and Forensic Soundness**

### **6.1 Integrity and Chain of Custody**

SentinelWire™ ensures forensic integrity by:

- Operating on read-only PCAP files

- Preserving original evidence without modification
- Generating timestamped, hash-verifiable outputs

## **6.2 Role in Legal Proceedings**

Outputs serve as expert analytical aids—not primary evidence. When supported by analyst testimony, they can:

- Reconstruct incident timelines
- Substantiate expert opinions
- Strengthen investigative narratives in legal and regulatory contexts

## **7. Behavioral Detection as a Legal Advantage**

Version v1.1 introduces advanced behavioral analysis capabilities:

- Time-Based Beaconing Detection – Identifies automated C2 communication patterns
- Lateral Movement Detection – Flags internal movement via SMB, LLMNR, and NBNS

Behavior-based indicators are inherently more defensible than static signatures, as they rely on observable network activity rather than assumptions about malicious intent.

## **8. Risk Management and Misuse Considerations**

SentinelWire™ is strictly defensive and does not perform:

- Active scanning
- Exploitation
- Unauthorized interception
- Payload manipulation

The human-in-the-loop model ensures clear responsibility attribution, reducing organizational and legal risk.

## **9. Strategic and Commercial Implications**

From a procurement and governance perspective, SentinelWire™ offers:

- **Reduced legal and compliance uncertainty**
- **High transparency and audit readiness**
- **Low regulatory friction** due to its privacy-by-design approach

Accompanying legal documentation positions SentinelWire™ as a compliance-aware investigative framework, suitable for regulated and high-risk environments.

## **10. Conclusion**

SentinelWire™ v1.1 delivers a legally robust, privacy-conscious, and evidentiary-sound approach to network investigation.

By emphasizing explainability, behavioral analysis, human oversight, and standardized classification, it provides a defensible alternative to opaque, automated detection systems enabling effective cybersecurity operations that also meet stringent legal and regulatory expectations.

## **I. Introduction**

The contemporary cybersecurity landscape is characterized by an increasing reliance on automated network-based threat detection, a shift that has precipitated fundamental legal inquiries regarding data protection, evidentiary reliability, and algorithmic accountability. In an environment where many prevailing cybersecurity tools function as opaque, automated detection systems, SentinelWire™ v1.1 distinguishes itself by adopting a fundamentally divergent operational philosophy: it is engineered as a human-in-the-loop, explainable, and method-driven investigation framework rooted in Wireshark and standardized threat modeling. This report proffers a doctrinal and regulatory analysis of the framework, rigorously evaluating its legal nature, compliance posture, and evidentiary implications within the context of European and international cybersecurity law.

By prioritizing transparency, proportionality, and judicial defensibility, the framework's design addresses critical lacunae in the current landscape of digital forensic tools, ensuring that technical efficacy does not come at the expense of legal compliance.<sup>1</sup>

## **II. Legal Characterization: Tool vs. Investigative Method**

### **A. SentinelWire™ as a Methodological Framework**

From a strictly legal standpoint, it is insufficient to characterize SentinelWire™ merely as a “software product”. Rather, it constitutes a comprehensive, structured investigative methodology that integrates analyst-controlled packet capture, automated indicator extraction, behavior-based scoring, and standardized threat classification into a cohesive workflow.<sup>2</sup> This distinction is paramount because courts and regulatory bodies traditionally assess the methodology used to derive conclusions rather than the underlying technical tools in isolation. SentinelWire™ aligns with this jurisprudential principle by ensuring that every analytical output remains traceable to observable packet-level data, thereby facilitating the rigorous peer review and judicial scrutiny required in legal proceedings. A critical nuance of this framework is that SentinelWire™ does not “decide” in an autonomous capacity; instead, it is designed explicitly to support human judgment.

### **B. Explainability and Accountability**

Modern legal frameworks are increasingly rejecting opaque algorithmic systems that produce non-explainable outcomes.<sup>3</sup> SentinelWire™ mitigates this legal risk by design, ensuring that every Indicator of Compromise (IOC) score, beaconing assessment, and lateral movement indicator is derived from explicit, inspectable criteria documented thoroughly within its rule sets and scoring algorithms. This architectural choice is particularly salient under European legal doctrines concerning algorithmic accountability and procedural fairness, as articulated in

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<sup>1</sup> Council of Europe, *Guidelines on the responsible use of artificial intelligence in the judiciary*, 2023.

<sup>2</sup> E. Casey, *Digital Evidence and Computer Crime*, 3rd ed., Academic Press, 2011; *R v. Tang* [2006] NSWCCA 167.

<sup>3</sup> Pasquale, F., *The Black Box Society*, Harvard University Press, 2015.

the proposed EU Artificial Intelligence Act and existing consumer protection statutes.<sup>4</sup> Because the framework’s outputs are inherently auditable, they satisfy the “right to an explanation “ in contexts where automated decisions might affect individuals, ensuring that the logic behind a security alert is transparent to the analyst and, by extension, the court.<sup>5</sup>

| Feature               | “Black Box “Automation                                  | SentinelWire™ Framework                                          |
|-----------------------|---------------------------------------------------------|------------------------------------------------------------------|
| <b>Decision Logic</b> | Opaque neural networks or hidden weights.               | Explicit, inspectable criteria and rule sets.                    |
| <b>Auditability</b>   | Difficult to reverse-engineer specific decisions.       | Inherently auditable; outputs map to raw data.                   |
| <b>Legal Standing</b> | Vulnerable to challenges regarding procedural fairness. | Satisfies “procedural fairness” and explainability requirements. |

### III. Data Protection and Privacy Law Analysis

#### A. Nature of Processed Data

The SentinelWire™ framework processes specific categories of network metadata, including IP addresses, DNS queries, TLS Server Name Indication (SNI), and protocol timing and frequency. It is critical to note that the system does *not* decrypt encrypted payloads nor inspect application content, a limitation that significantly frames its privacy impact. Under the jurisprudence of the General Data Protection Regulation (GDPR), such data may constitute personal data only insofar as it is linkable to an identifiable natural person.<sup>6</sup> While the Court of Justice of the European Union (CJEU) held in *Breyer v. Germany* that dynamic IP addresses can be personal data when an operator possesses additional information required for identification, SentinelWire™ typically operates in an organizational context where such direct linkage is absent during the initial triage phase, thereby reducing immediate privacy impacts.<sup>7</sup>

#### B. Lawful Basis and Proportionality under GDPR

The framework supports compliance through three primary lawful bases:

<sup>4</sup> Proposal for a Regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), COM/2021/206 final; Directive 2005/29/EC on unfair commercial practices.

<sup>5</sup> Goodman, B. & Flaxman, S., "European Union Regulations on Algorithmic Decision-Making and a ‘Right to Explanation’," *AI Magazine*, 2017.

<sup>6</sup> Regulation (EU) 2016/679 (GDPR), Article 4(1).

<sup>7</sup> CJEU, *Breyer v. Germany*, Case C-582/14.

1. **Legitimate Interests (Art. 6(1)(f)):** The necessity of securing networks generally outweighs minimal residual privacy impacts, provided the processing is proportionate.
2. **Legal Obligation (Art. 6(1)(c)):** For entities under the **NIS2 Directive**, network monitoring is a mandatory security measure.
3. **Security of Processing (Art. 32):** The GDPR imposes a direct obligation to ensure system integrity and resilience, which SentinelWire™ actively supports.

The deployment of SentinelWire™ is supported by several lawful bases under the GDPR. Primarily, Article 6(1)(f) GDPR allows processing based on legitimate interests in securing networks and assets, while Article 6(1)(c) GDPR is relevant for compliance with legal security obligations, such as those mandated by NIS2.

Furthermore, Article 32 GDPR imposes a direct obligation to ensure the ongoing confidentiality, integrity, and resilience of processing systems, which this framework actively supports. A Legitimate Interests Assessment (LIA) for utilizing this framework would likely conclude that the necessity for security, combined with the limited scope of processing and the safeguards built into the methodology, outweighs any minimal residual impact on individual privacy.<sup>8</sup> Crucially, the framework's recommendation of short, 30–60 second capture windows and purpose-limited analysis significantly strengthens proportionality arguments under Article 5(1)(b) GDPR.<sup>9</sup>

### **C. Data Minimization, Storage Limitation, and E-Privacy Considerations**

SentinelWire™ operationalizes the GDPR principle of data minimization (Article 5(1)(c)) by strictly avoiding long-term, pervasive traffic recording. Instead, it limits capture scope to targeted triage windows based on specific alerts or suspicions and produces derivative analytical outputs—such as scores, classifications, and reports—rather than retaining large volumes of raw packet data. This approach materially reduces legal exposure compared to persistent monitoring systems and aligns with the principle of storage limitation (Article 5(1)(e)), as raw captures can be securely deleted post-analysis without losing the analytical record.<sup>10</sup>

Additionally, while the ePrivacy Directive (2002/58/EC) strictly regulates the interception of communications, its legitimate business practice exemption and provisions for network security are applicable.<sup>11</sup> SentinelWire™, as a defensive tool analyzing traffic on an organization's own network for security purposes, falls within recognized exceptions for activities necessary to ensure the security of the service.<sup>12</sup>

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<sup>8</sup> European Data Protection Board (EDPB), *Guidelines 2/2019 on the processing of personal data under Article 6(1)(f)*, 2019.

<sup>9</sup> EDPB, *Guidelines 4/2019 on Article 6(1)(b) GDPR*.

<sup>10</sup> Voigt & von dem Bussche, *The EU General Data Protection Regulation (GDPR)*, Springer, 2017.

<sup>11</sup> Directive 2002/58/EC (ePrivacy Directive), Article 5(1) & Recital 20.

<sup>12</sup> Article 29 Data Protection Working Party, *Opinion 2/2017 on data processing at work*, WP 249, 2017.

## **IV. Cybersecurity Regulatory Frameworks**

### **A. NIS2 Directive Compliance**

Under the European Union's NIS2 Directive, regulated entities are mandated to implement appropriate technical and organizational measures for incident detection, handling, and reporting.<sup>13</sup> SentinelWire™ directly supports these obligations by enabling early-stage incident detection via behavioral anomalies, structured incident analysis with documented procedures, and the generation of clear, standardized outputs for internal reporting and potential regulatory submission. Consequently, the framework assists organizations in meeting NIS2 requirements for baseline security measures (Article 21) and incident handling (Article 23).<sup>14</sup>

### **B. Alignment with ISO/IEC 27001 & 27002 and ENISA Guidelines**

Furthermore, SentinelWire™ contributes to compliance with key controls found in ISO/IEC 27002:2022. Specifically, it supports Control A.8.15 (Logging) and Control A.8.16 (Monitoring activities) by providing a structured mechanism for network observation. It also aids in Information Security Incident Management Planning and Preparation (Control A.5.24) and the Management of Technical Vulnerabilities (Control A.8.8). The framework's documentation-centric design is particularly valuable for audits and post-incident reviews, providing concrete evidence of a systematic approach to threat detection.<sup>15</sup> This aligns with principles outlined by the European Union Agency for Cybersecurity (ENISA) for incident analysis and digital forensics, specifically the emphasis on methodology, traceability, and the use of standardized taxonomies.<sup>16</sup>

## **V. Evidentiary Value and Digital Forensics**

### **A. Chain of Custody and Integrity**

To preserve forensic integrity, SentinelWire™ operates exclusively on read-only PCAP files, ensuring that original evidence is never altered during the analytical process. It avoids packet modification during analysis and produces timestamped, cryptographically hash-verifiable outputs for each analytical run, while simultaneously maintaining detailed activity logs of the analyst's actions within the framework. These features align with established digital evidence best practices, such as those outlined in the ACPO Good Practice Guide for Digital Evidence and ISO/IEC 27037:2012 guidelines for the identification, collection, acquisition, and preservation of digital evidence.<sup>17</sup>

### **B. Admissibility and the Role of Expert Evidence**

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<sup>13</sup> Directive (EU) 2022/2555 (NIS2), Articles 20-21.

<sup>14</sup> ENISA, *Technical Guidance on Incident Reporting*, 2023.

<sup>15</sup> ISO/IEC 27002:2022, Information security, cybersecurity and privacy protection — Information security controls.

<sup>16</sup> ENISA, *Good Practice Guide on Incident Management*, 2022.

<sup>17</sup> ISO/IEC 27037:2012, Guidelines for identification, collection, acquisition, and preservation of digital evidence; ACPO, *Good Practice Guide for Digital Evidence*, 2012.

SentinelWire™ outputs should be understood legally as expert analytical aids, not standalone proof. Courts routinely admit such analytical summaries when standards such as *Daubert* or *Frye* (in the U.S.) or their national equivalents are met.<sup>18</sup> The framework supports admissibility because its methodology is tested, peer-reviewed, and has a known error rate; the analysis is based on sufficient facts or data; and the technique utilizes generally accepted scientific principles, such as packet analysis and the MITRE ATT&CK framework. The analyst’s ability to testify to the process, explain the rules applied, and demonstrate the causal link between raw data and conclusions is paramount to its evidentiary weight.<sup>19</sup>

### **C. MITRE ATT&CK as a Standardized Interpretive Layer**

The integration of the MITRE ATT&CK framework provides a shared, peer-reviewed vocabulary for describing adversary behavior. This standardization significantly enhances inter-organizational communication during incident response and improves regulatory reporting clarity. From a legal perspective, MITRE functions as a translation layer, converting technical observables into a narrative of tactics, techniques, and procedures (TTPs) that are comprehensible to non-technical judges, juries, and regulators, thereby reducing ambiguity and the risk of misinterpretation in legal proceedings.<sup>20</sup>

## **VI. Risk of Misuse, Liability, and Commercial Implications**

### **A. Defensive Nature and Human Responsibility**

SentinelWire™ is strictly defensive; it does **not** perform active scanning of external networks, vulnerability exploitation, or payload manipulation. Accordingly, it falls squarely within defensive cybersecurity tooling, limiting exposure under criminal misuse statutes such as the U.S. Computer Fraud and Abuse Act (CFAA) or Articles 2-6 of the Council of Europe Budapest Convention on Cybercrime.<sup>21</sup> By requiring analyst interpretation at every critical stage—from capture trigger to final classification—SentinelWire™ supports a clear model of responsibility attribution. This “human-in-the-loop” model is a critical factor in civil liability assessments and administrative proceedings, ensuring that a human remains accountable for the investigative conclusions drawn.<sup>22</sup>

### **B. Commercial, Licensing, and Dual-Use Considerations**

While any network analysis tool may theoretically be repurposed, SentinelWire™’s design and documentation strongly anchor it within defensive and investigative operations. Its lack of offensive capabilities reduces its classification as “intrusion software” under export control

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<sup>18</sup> *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993); *Frye v. United States*, 293 F. 1013 (D.C. Cir. 1923).

<sup>19</sup> *R v. M(N)* [2017] EWCA Crim 2165 (UK).

<sup>20</sup> BE Strom, MITRE, *MITRE ATT&CK® Design and Philosophy*, 2020.

<sup>21</sup> Convention on Cybercrime (Budapest Convention), ETS No. 185; 18 U.S.C. § 1030 (U.S. CFAA).

<sup>22</sup> European Commission, *Ethics Guidelines for Trustworthy AI*, 2019.

regimes like the EU Dual-Use Regulation (EU 2021/821) and the Wassenaar Arrangement, simplifying procurement for most entities.<sup>23</sup>

This legal assessment itself functions as a risk-reduction instrument, demonstrating due diligence in tool selection and serving as a compliance assurance artifact for auditors and regulators. It may be leveraged as a standalone legal whitepaper or bundled with enterprise offerings to provide immediate clarity on the framework's legal and operational standing.

## **VII. Conclusion**

SentinelWire™ v1.1 represents a legally defensible, privacy-conscious, and methodologically transparent approach to network investigation.

By emphasizing explainability over opacity, behavioral analysis over content inspection, and human oversight over full automation, it places itself well within contemporary legal and regulatory expectations for cybersecurity operations. By bridging the gap between technical necessity and legal compliance, SentinelWire™ provides a robust foundation for incident response that is capable of withstanding rigorous judicial and regulatory scrutiny.

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<sup>23</sup> Regulation (EU) 2021/821 setting up a Union regime for the control of exports, brokering, technical assistance, transit and transfer of dual-use items; The Wassenaar Arrangement on Export Controls for Conventional Arms and Dual-Use Goods and Technologies.

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Voigt & von dem Bussche, *The EU General Data Protection Regulation (GDPR)*, Springer, 2017.



## About SentinelWire™

A Wireshark-driven, analyst-controlled network investigation toolkit that enables legally defensible, privacy-conscious, and regulatorily compliant methodologies to cybersecurity operations.

### Legal and Regulatory Focus

- Cybersecurity law (NIS2, ISO/IEC 27001),
- Privacy regulations (GDPR, CJEU case law)
- Admissibility and digital forensics

### Network Investigation Capabilities

- Beacons detection (automated C2 patterns).
- Lateral movement analysis (SMB, LLMNR, NBNS)
- Indicator of Compromise intelligence.

December 2025